

Engineering Probability

Homework 7

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October 21st 2025

(1) Gaussian Temperature in Antarctica.

The peak temperature T (in °F) on a July day in Antarctica is a Gaussian random variable with variance 225. With probability $1/2$, the temperature T exceeds 10°F .

$$\mu = 10, \sigma = \sqrt{225}$$

- (a) Compute $P[T > 32]$, the probability the temperature is above freezing.

$$P[T > 32] = Q\left(\frac{32 - 10}{\sqrt{225}}\right) = 1 - \Phi\left(\frac{22}{15}\right) \approx 1 - \Phi(1.47) = 0.0708$$

- (b) Compute $P[T < 0]$.

$$P[T < 0] = \Phi\left(\frac{0 - 10}{\sqrt{225}}\right) = 1 - \Phi\left(\frac{10}{15}\right) \approx 1 - \Phi(0.67) = 0.2514$$

- (c) Compute $P[T > 60]$.

$$P[T > 60] = Q\left(\frac{60 - 10}{\sqrt{225}}\right) = 1 - \Phi\left(\frac{50}{15}\right) \approx 1 - \Phi(3.33) = 0.0005$$

(2) Blackboard Error Payouts (Gaussian Approximation).

A professor pays \$0.25 for each blackboard error made in lecture to the student who points out the error. Over n years, the total amount paid (in dollars) can be approximated by a Gaussian random variable Y_n with $\mathbb{E}[Y_n] = 40n$ and $\text{Var}[Y_n] = 100n$.

$$\mu = 40n, \sigma = \sqrt{100n}$$

(a) What is $P[Y_{20} > 1000]$?

$$P[Y_{20} > 1000] = Q\left(\frac{1000 - 40(20)}{\sqrt{100(20)}}\right) \approx Q(4.47) = 3.91 \cdot 10^{-6}$$

(b) How many years n are required so that $P[Y_n > 1000] > 0.99$?

$$P[Y_n > 1000] = 0.99, Q(z) = 0.99, -2.32 < z < -2.33$$

$$-2.32 = \frac{1000 - 40(n)}{\sqrt{100(n)}}, n \approx 28.073$$

$$-2.33 = \frac{1000 - 40(n)}{\sqrt{100(n)}}, n \approx 28.087$$

$$\therefore n = 29$$

(3) Shot Put Distance as a Mixture.

With probability 0.7, the toss distance is $D = 60 + X$ meters, where X is exponential with mean $\mu = 10$. Otherwise (probability 0.3), a foul is committed and $D = 0$.

(a) Find the CDF $F_D(d)$ of D .

$$\int_{60}^d 0.7 \cdot \frac{1}{10} e^{\frac{-(d-60)}{10}} dd = 0.7 \left(-e^{\frac{-(d-60)}{10}} \right) \Big|_{60}^d = 0.7 \left(1 - \left(e^{\frac{-(d-60)}{10}} \right) \right)$$
$$F_D(d) = u(d)0.3 + 0.7 \left(1 - e^{\frac{-(d-60)}{10}} \right) u(d-60)$$

(b) Find the PDF $f_D(d)$ of D .

$$f_X(x) = \frac{1}{\mu} e^{\frac{-x}{\mu}}, \{x > 0\}$$
$$f_D(d) = 0.3\delta(d) + 0.7 \cdot \frac{1}{10} e^{\frac{-(d-60)}{10}} u(d)$$

(4) Impulse + Exponential Tail (Mixed PDF).

Let X be a mixed random variable with an atom at $x = 0$ and a continuous exponential tail:

$$f_X(x) = p\delta(x) + (1-p)\lambda e^{-\lambda x}u(x), \quad p \in (0, 1), \lambda > 0,$$

where $\delta(\cdot)$ is the unit impulse (delta function) and $u(\cdot)$ is the unit step.

(a) Find the CDF $F_X(x)$ for all $x \in R$ and clearly indicate the jump at $x = 0$.

$$\int_0^x (1-p)\lambda e^{-\lambda x} = -(1-p)e^{-\lambda x} \Big|_0^x = -(1-p)e^{-\lambda x} - -(1-p) = (1-p)(1 - e^{-\lambda x})$$
$$F_X(x) = p\delta(x) + (1 - (1-p)e^{-\lambda x}) u(x)$$

Verify that F_X is right-continuous and that $\lim_{x \rightarrow -\infty} F_X(x) = 0$, $\lim_{x \rightarrow \infty} F_X(x) = 1$.

$$\lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\lim_{x \rightarrow \infty} F_X(x) = \lim_{x \rightarrow \infty} 1 - (1-p)e^{-\lambda(x)} = 1 - (1-p)(0) = 1$$

- (b) Compute $P[X = 0]$, $P[X > t]$ for $t \geq 0$, and the first two moments $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$. Hence, find $\text{Var}[X]$. (Show all steps. You may use $\int_0^\infty x^k e^{-\lambda x} dx = \frac{k!}{\lambda^{k+1}}$ for $k = 0, 1, 2$.)

$$P[X = 0] = p$$

$$P[X > t] = (1 - p)e^{-\lambda t}$$

$$\mathbb{E}[X] = 0 \cdot p + \int_0^\infty (x(1 - p)\lambda e^{-\lambda x}) dx = \lambda(1 - p) \frac{1!}{\lambda^2} = \frac{1 - p}{\lambda}$$

$$\mathbb{E}[X^2] = 0^2 \cdot p + \int_0^\infty (x^2(1 - p)\lambda e^{-\lambda x}) dx = \lambda(1 - p) \frac{2!}{\lambda^3} = \frac{2 - 2p}{\lambda^2}$$

$$\text{Var}[X] = \frac{2 - 2p}{\lambda^2} - \left(\frac{1 - p}{\lambda} \right)^2 = \frac{1 - p^2}{\lambda^2}$$

(5) Linear and Nonlinear Transforms of a Mixed RV.

Let X be as in Question 4. Define $Y = aX + b$ with $a > 0$ and $b \in \mathbb{R}$; and define $Z = X^2$.

- (a) Derive the PDF of Y in the unified (impulse + continuous) form. Clearly specify the location and the weight of the impulse, and give the continuous part of $f_Y(y)$ for $y > b$.

$$f_X(x) = p\delta(x) + (1-p)\lambda e^{-\lambda x}u(x), \quad p \in (0,1), \lambda > 0,$$

$$F_X(x) = \left(1 - (1-p)e^{-\lambda x}\right)u(x)$$

$$F_Y(y) = \left(1 - (1-p)e^{-\lambda\left(\frac{y-b}{a}\right)}\right)u\left(\frac{y-b}{a}\right)$$

$$\frac{dF_Y(y)}{dy} = \left(\frac{\lambda}{a}(1-p)e^{-\lambda\left(\frac{y-b}{a}\right)}\right)u\left(\frac{y-b}{a}\right) + \left(1 - (1-p)e^{-\lambda\left(\frac{y-b}{a}\right)}\right)\delta\left(\frac{y-b}{a}\right)$$

$$\frac{dF_Y(y)}{dy} = f_Y(y) = \frac{\lambda}{a}(1-p)e^{-\lambda\left(\frac{y-b}{a}\right)}u(y-b) + p\delta(y-b)$$

Weight of impluse at $(y = 0)$ is p

$$f_Y(y) = \frac{1}{a} \cdot \lambda(1-p)e^{-\lambda\left(\frac{y-b}{a}\right)}, y > b$$

- (b) Derive the PDF of Z . Give it in unified form and identify any impulses (location and weight). Then provide the explicit formula for the continuous part $f_Z(z)$ for $z > 0$.

$$F_X(x) = \left(1 - (1-p)e^{-\lambda x}\right)u(x)$$

$$F_Z(z) = \left(1 - (1-p)e^{-\lambda\sqrt{z}}\right)u(\sqrt{z})$$

$$\frac{dF_Z(z)}{dz} = (1-p) \left(\frac{\lambda}{2\sqrt{z}}\right) e^{-\lambda\sqrt{z}}u(\sqrt{z}) + \left(1 - (1-p)e^{-\lambda\sqrt{z}}\right)\delta(\sqrt{z})$$

$$\frac{dF_Z(z)}{dz} = f_Z(z) = (1-p) \left(\frac{\lambda}{2\sqrt{z}}\right) e^{-\lambda\sqrt{z}}u(\sqrt{z}) + p\delta(z)$$

Weight of impluse at $(z = 0)$ is p

$$f_Z(z) = (1-p) \left(\frac{\lambda}{2\sqrt{z}}\right) e^{-\lambda\sqrt{z}}, z > 0$$

(6) Power Through a Resistor.

The voltage X across a $1\ \Omega$ resistor is $\text{Uniform}(0, 1)$. The instantaneous power is $Y = X^2$.

(a) Find the CDF $F_Y(y)$.

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x, & 0 < x < 1 \\ 1, & x > 1 \end{cases}$$

$$X = \sqrt{Y}$$

$$F_Y(y) = \begin{cases} 0, & y < 0 \\ \sqrt{y}, & 0 < y < 1 \\ 1, & y > 1 \end{cases}$$

(b) Find the PDF $f_Y(y)$.

$$f_Y(y) = \begin{cases} \frac{1}{2\sqrt{y}}, & 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

(7) Square of an Exponential.

Let $Y = X^2$ where X has PDF

$$f_X(x) = \begin{cases} 9e^{-9x}, & x \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the PDF $f_Y(y)$.

$$f_Y(y) = \frac{f_X(\sqrt{y})}{\left|\frac{d}{dx}x^2\right|} + \frac{f_X(-\sqrt{y})}{\left|\frac{d}{dx}(x^2)\right|} = f_X(\sqrt{y}) \cdot \frac{1}{|2x|} = \frac{9e^{-9\sqrt{y}}}{2\sqrt{y}}$$

$X = \sqrt{Y}$
 $0, \{x < 0\}$

Canceled second term f_X is only defined when x is negative, which is outside of the domain of X , or otherwise 0.

(b) Compute $\mathbb{E}[Y]$.

$$E[Y] = E[X^2] = \frac{2}{(\lambda)^2} = \frac{2}{(9)^2}$$

(c) Compute $\text{Var}[Y]$.

$$E[Y^2] = E[X^4] = \frac{4!}{\lambda^4} = \frac{4!}{9^4}$$
$$\text{Var}[Y] = \frac{4!}{9^4} - \left(\frac{2}{(9)^2}\right)^2 = \frac{20}{9^4}$$